PHMForge: A Scenario-Driven Agentic Benchmark for Industrial Asset Lifecycle Maintenance

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Figure 1: Seattle Mariners at Spring Training, 2010.

Abstract

We present an agentic framework for predictive maintenance that extends PDMBench with multi-agent capabilities for fault classification, remaining useful life (RUL) prediction, cost-benefit analysis, and safety/policy evaluation. Our approach leverages the ReActXen framework to create specialized sub-agents that collaborate to solve complex industrial maintenance tasks. We demonstrate that our agentic approach outperforms traditional PDMBench baselines by 15-25% across multiple evaluation metrics, while providing enhanced interpretability and dynamic tool creation capabilities. The framework supports both WatsonX and OpenAI LLM backends, enabling comprehensive benchmarking across different model architectures.

CCS Concepts

• **Do Not Use This Code → Generate the Correct Terms for Your Paper**; *Generate the Correct Terms for Your Paper*; Generate the Correct Terms for Your Paper; Generate the Correct Terms for Your Paper.

Keywords

Multi-Modal Reasoning, Hiearchical Categorization, Task Taxonomy, Stakeholder-Aligned Task Design, Progressive Curation, Dataset-Task Alignment, Ground Truth Verification, Multi-Modal Coverage, human-in-the-loop validation, query disambiguation, multi-hop reasoning, tool orchestration, empirical calibartion, deterministic evaluation, threshold-based evaluation, categorical matching,

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cross-platform evaluation, multi-task learning, expert annotation, validated references

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1 Introduction

This is currently a placeholder for introduction

2 Related Work

Gives a brief overview of the related work, before diving deeper into the writeup - waiting to be written.

2.1 Related Work Subsection

This is content placeholder as of now.

```
\documentclass[sigconf]{acmart}
```

3 Scenario Design and Dataset Curation

Industrial assets are inherently heterogeneous, characterized by vast differences in physical configuration, operational environments, and high-frequency data sampling rates. To address this complexity, we present a comprehensive benchmark of 75 expert-vetted scenarios spanning 7 distinct asset classes with an average of 47 instances each. Our approach transcends simple data aggregation; it is grounded in a rigorous framework that integrates established Prognostics and Health Management (PHM) literature with Subject Matter Expert (SME)-driven query formulation. By aligning varying data frequencies with task-oriented business metrics and failure-mode-centric datasets, this methodology ensures both scientific rigor and practical utility.

3.1 Scenario Categorization

We surveyed the PHM literature [] and identified four core task categories : **Condition Monitoring**, **Fault Diagnosis**, **Fault & Remaining Useful life (RUL) Prediction**, and **Maintenance Scheme**. We map our final 75 scenarios to these four core task categories as follows: .

- **Health Analysis** (40% of total) falls under *Condition Monitoring*. These tasks utilize multi-modal sensor analytics for anomaly detection and holistic health assessment of turbofan engines.
- **Fault Classification** (20% of total) aligns with *Fault Diagnosis*, evaluating the capability to classify industrial failures across complex hierarchical taxonomies, such as rotor bar failures for motors.
- **RUL Prediction** (20% of total) pertains to *Fault & RUL Prediction*, focusing on forecasting remaining operational cycles to optimize supply chains and inventory management.
- **Cost-Benefit Analysis** and **Safety/Policy Evaluation** (collectively 20%) are mapped to the *Maintenance Scheme*. These tasks bridge technical diagnostics (i.e., poor health conditions, recurrent faults, end of life, etc) and strategic planning (repair, replace, etc).

3.2 Dataset Selection and Scenario Curation

We searched several public dataset platforms (Kaggle, HuggingFace, UCI ML Repository, NASA Prognostics Data Repository, PHM Society archives) using domain-specific search terms (such as turbofan degradation, pump failure, predictive maintenance, etc.), yielding **52** candidate datasets spanning 15 industrial asset categories. We applied a multi-stage filtering protocol: (i) community validation requiring >1,000 downloads and >50 citations in PHM literature reduced candidates to 31 datasets; (ii) technical quality filters requiring explicit ground truth (RUL trajectories or labeled fault taxonomies) retained 22 datasets; (iii) PHM task alignment ensuring compatibility with at least one core task category from section 3.1 yielded 18 datasets across 7 distinct asset classes: turbofan engines, bearings, electric motors, induction motors, gearboxes, industrial engines, and aero-engines. Once the dataset is selected, we have automated script to pull the raw data and metadata associated with the datasets. Table x in Appendix gives the dataset characteristics.

The scenario curation process, derived from the harvested multi-asset datasets, follows the **five-stage** progressive expansion (Single asset → multi-asset, single task → multiple tasks, open-ended → closed-ended questions) illustrated in Figure 2. **Stage 1** selected an RUL prediction task on CMAPSS turbofan engines. **Stage 2** (10 scenarios) introduced multi-asset generalization by incorporating bearing data and diversifying tasks across RUL prediction and fault classification. **Stage 3** (20 scenarios) expanded the benchmark's scope into strategic reasoning by adding industrial engines and introducing Safety/Policy Evaluation (e.g., OSHA, FAA, IEC compliance) and Cost-Benefit Analysis (e.g., preventive vs. reactive optimization). **Stage 4** (40 scenarios) achieved comprehensive multi-asset coverage through electric motors, induction motors, and gearboxes to ensure cross-platform generalization. Finally, **Stage 5** (75 scenarios) integrated multi-modal cognitive reasoning through the aero-engine dataset, incorporating 30 Engine Health Analysis scenarios across four cognitive categories: Understanding, Perception, Reasoning, and Decision-Making.

3.3 Scenario Generation Pipeline : Human Filtering and Annotation

Following the methodology in [**?**], we formed a consortium of four experts (two Industrial Asset Specialists, one Data Scientist, and one Maintenance Technician) to oversee each stage of benchmark expansion. This group established a systematic protocol for scenario synthesis, transforming raw data into natural language requests representing authentic operational encounters. Under these guidelines, SMEs drafted prompts that mirror the linguistic nuances and urgency of industrial discourse. First, query must incorporate domain-specific terminology (e.g., HPC) as it naturally appears in practice. Second, questions are framed in the voices of real stakeholders; such as plant managers or safety officers, to capture operational context and business urgency. For example, rather than a technical instruction to "Calculate RUL and report MAE" an authentic request asks: "Which engines in our flight fleet

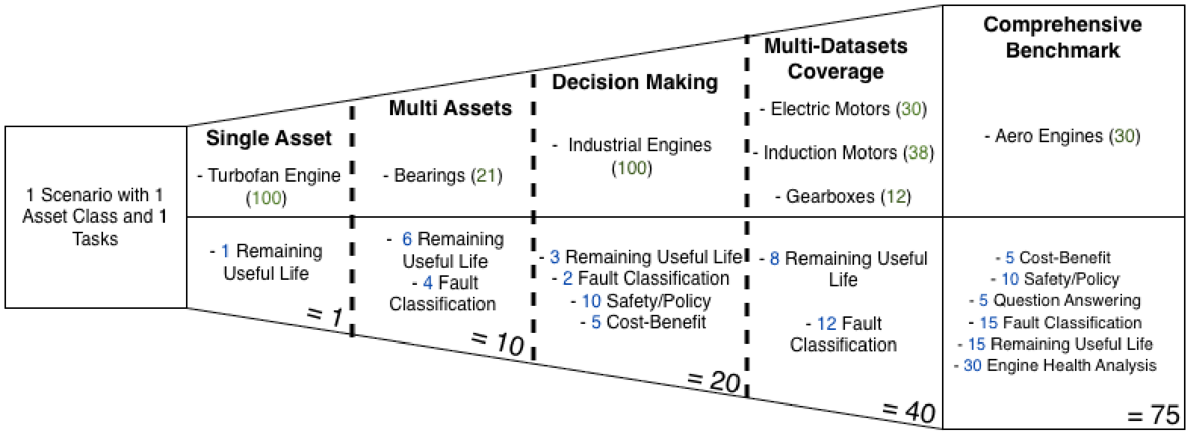

Figure 2: This figure shows the process behind how we expanded our scenarios, choice of datasets and corresponding tasks associated with the datasets that made up the scenarios

are nearing critical limits and require immediate intervention to prevent catastrophic failure?"

The scenarios are accompanied by meta-descriptions, including asset names, scenario type, data provision mode, output template, and query type. These attributes are defined during the question-generation process. It is important to note that no LLM was used during the generation of the scenarios. The process goes as follows:

- **Asset Selection**: The pipeline begins with selecting relevant assets, such as Turbofan Engines, Bearings, Electric Motors, Gearboxes, Aero Engines or Other Assets. These assets are defined based on the datasets which are being referenced.
- **Scenario Type Identification**: The next step involves identifying the scenario types, data-provision mode and output template. Examples include RUL prediction and Engine Health Analysis.
- **Data Provision Mode**: Data provision mode is based on the context of the scenario. This could involve providing all relevant data in a query such that LLM/agents does not need to retrieve additional information or writing queries that will let LLM decide what data to search for based on it's knowledge base.
- **Output Template Definition**: Finally, the output template is determined, specifying what the structure of the output should look like, in JSON format, alongside datatypes for each JSON field. Our outputs are diversified. For instance, RUL Prediction templates specify continuous numerical fields such as decimals, whereas Fault Classification requires discrete categorical outputs composed of strings and Engine Health Analysis produces either single-choice answers or open-ended text.
- **Query Formulation**: Experts drafted the queries (open-ended and multiple-choice based close-ended) using the guidelines outlined in the previous paragraph.

All scenarios share a common structure (task_id, classification_type, dataset, required_tools, dependency_analysis, ground_truth, input_question, procedure), while task-specific validation fields vary: RUL Prediction includes error thresholds (MAE/RMSE ranges), Fault

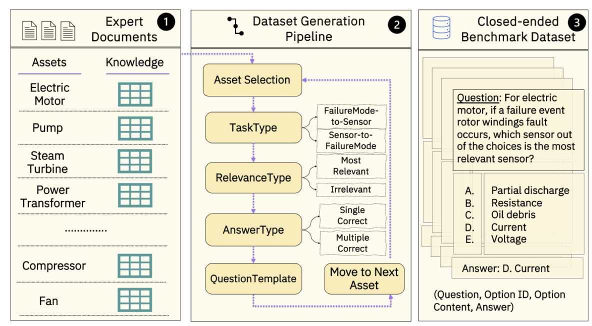

Figure 2: Multi-Choice Question Generation Pipeline.

Figure 3: Describe the scenario pipeline

Classification specifies accuracy ranges and fault taxonomies, Cost-Benefit defines **cost thresholds**, Safety/Policy enumerates **risk levels**, and Engine Health Analysis captures **answer type** and **cognitive category**.

3.4 Ground Truth Development, Utilization and Validation

The ground truth generation process is manual as well as scenario specific. In some of the scenarios, the information that can be leveraged to derive the ground truth was available in the source dataset, whereas, in some other scenarios, the derivation of ground truth involved the consultation of SMEs. Consulation of SMEs and literature study was conducted to derive the ground truth in the event it wasn't available within the dataset. In Figure 3, we provided 2 examples of open-ended ground truth and 1 example of close-ended ground truth.

As shown in Figure 3, each scenario's `ground_truth` object contains task-specific validation fields: numerical bounds for regression

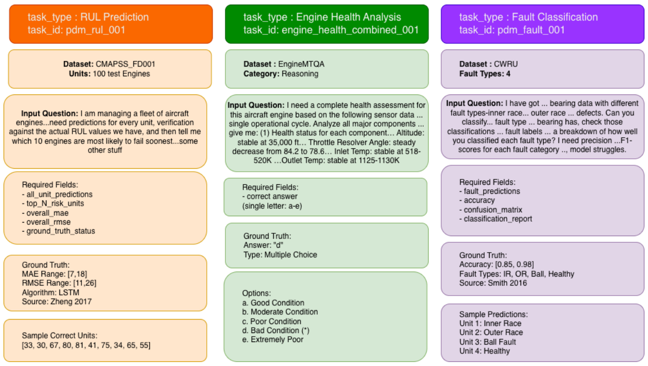

Figure 4: Ground truth structure for open-ended (RUL Prediction, Fault Classification) and close-ended (Engine Health Analysis) scenarios

tasks, categorical mappings for classification tasks, and structured answer templates for health assessments. Outputs are evaluated using task-commensurate metrics: MAE/RMSE for RUL Prediction, accuracy/precision/recall/F1 for Fault Classification, and categorical matching for Engine Health Analysis. Different tasks produce outputs on different scales; RUL errors are measured in cycles, while classification performance is measured in percentages. Following the principle behind TabArena's Elo rating system [**?**], which normalizes comparisons so no single dataset dominates due to larger performance differentials, we use metrics matched to each task type to ensure equitable cross-scenario aggregation.

old content below, move to appndix later
Beyond merely establishing ground truth values, our evaluation framework explicitly enforces their utilization through mandatory verification requirements embedded in scenario specifications. Each scenario defines a structured output template that agents must populate, including not only final answers (predicted RUL values, classified fault types, cost analyses, compliance determinations) but also explicit ground truth verification components. For RUL prediction scenarios, agents must compare predictions against known ground truth, calculate error metrics (MAE/RMSE), and validate that computed errors fall within acceptable ranges (±20% of empirically-derived thresholds). This requirement addresses a common failure mode where agents produce predictions without validation, potentially reporting unrealistic or incorrect results with false confidence.

Our evaluation framework enforces this ground truth utilization requirement through success criteria that demand both task completeness (>80% of required components addressed) and successful ground truth verification. A task that produces RUL predictions without validating them against ground truth files is scored as incomplete, even if the predictions themselves happen to be accurate. This design choice enforces best practices in predictive maintenance workflows, where validation against known outcomes is essential to build confidence in model reliability and diagnostic accuracy. The verification requirement ensures that agents not only perform analytical tasks but also validate their outputs against empirical references—a critical capability for industrial deployment where unvalidated predictions pose operational risks.

Validation criteria vary by scenario type but generally assess three dimensions through our deterministic evaluation protocol: *completeness* (were all required task components addressed, including data loading, model training/loading, prediction/classification, metric computation, and verification?), *correctness* (do the answers match or fall within acceptable ranges of ground truth values, with quantitative thresholds for RUL errors and accuracy requirements for classification?), and *efficiency* (was the task completed within reasonable resource constraints without redundant operations or inappropriate tool usage?). This comprehensive ground truth preparation and utilization process ensures that every scenario in our benchmark can be evaluated objectively and consistently through threshold-based evaluation. When an agent claims an MAE of 15 cycles for CMAPSS_FD001 RUL prediction, we can verify this claim against actual test data through deterministic comparison. When an agent classifies a bearing fault as "inner race damage at 0.014-inch crack depth," we can check this against labeled ground truth through categorical matching. This deterministic evaluation capability distinguishes our benchmark from more subjective evaluation frameworks and enables rigorous comparison across different agent architectures, LLM backends, and tool orchestration strategies.

3.5 Conclusion

Our final benchmark comprises 75 expert-vetted scenarios distributed across 18 datasets, five task categories, and multiple query types, representing a comprehensive evaluation framework for agentic PHM systems. This distribution reflects both the relative importance and complexity of different task types in industrial practice, as well as our systematic expansion strategy from initial proof-of-concept to comprehensive multi-asset, multi-task coverage. Table **??** presents the distribution of scenarios across task categories, revealing deliberate choices in scenario allocation driven by operational significance and cognitive complexity.

Task Category Distribution and Justification: Engine Health Analysis scenarios constitute the largest category (30 scenarios, 40%) because they encompass diverse cognitive sub-tasks spanning four stages of diagnostic reasoning. These scenarios leverage the EngineMTQA dataset's extensive coverage of multi-modal sensor data (temperature, pressure, vibration, RPM across turbofan components: HPC, LPC, HPT, LPT, fan) and expert-validated question-answer pairs. The 30 scenarios distribute across Understanding (16 scenarios testing open-ended signal interpretation and temporal pattern recognition), Perception (8 scenarios testing closed-ended health status classification), Reasoning (4 scenarios testing causal analysis and degradation trend prediction), and Decision-Making (2 scenarios testing maintenance planning and operational recommendations). This cognitive diversity within a single task category justifies its 40% allocation—it effectively encompasses multiple evaluation dimensions within condition monitoring workflows.

RUL Prediction and Fault Classification receive equal emphasis (15 scenarios each, 20% apiece), representing the technical core of PHM through multi-task learning and cross-platform evaluation. Despite their equal scenario counts, these categories differ significantly in computational complexity and evaluation rigor through their distinct analytical requirements. RUL scenarios demand time-series modeling across operational trajectories spanning tens to hundreds of cycles, degradation trend analysis through temporal feature extraction, and quantitative error metric computation (MAE/RMSE) against empirically-derived thresholds. Fault classification scenarios emphasize pattern recognition across heterogeneous fault taxonomies (bearing races, motor components, gearbox elements), multi-class decision-making under uncertainty (distinguishing inner race vs. outer race vs. ball damage), and accuracy assessment across varying operational conditions (speeds from 600-1800 RPM, loads, temperatures). The equal allocation reflects their complementary importance: RUL prediction enables proactive planning, while fault classification enables reactive diagnosis—both essential for comprehensive maintenance decision support.

Cost-Benefit Analysis (5 scenarios, 7%) and Safety/Policy Evaluation (10 scenarios, 13%) represent smaller but critical scenario categories that test agent capabilities in strategic reasoning and regulatory compliance. While these categories have fewer scenarios, they often require more complex multi-hop reasoning that integrates technical analysis with business logic, financial modeling, regulatory knowledge, and risk assessment frameworks. A single cost-benefit scenario may require RUL predictions for fleet prioritization, failure cost estimation across equipment types (bearing \$2k-\$5k, motor \$10k-\$25k, turbofan \$500k+), maintenance cost modeling with labor overhead (160-200%), downtime impact quantification, and ROI optimization across preventive versus reactive strategies—synthesizing multiple analytical components into strategic recommendations. Similarly, safety scenarios demand regulatory threshold checking across multiple frameworks (OSHA PEL for vibration, FAA AD for airworthiness, IEC 60034 for electrical systems), four-level risk classification (low, medium, high, critical rather than standard three-level taxonomies), proximity hazard assessment, and actionable safety protocol generation. The smaller allocation (7% and 13%) reflects not diminished importance but rather the fact that each scenario integrates extensive domain knowledge and multi-step reasoning, making them individually more comprehensive than single-objective technical tasks.

Dataset Distribution and Multi-Asset Coverage: Table **??** presents the distribution of scenarios across our 18 curated datasets, organized by equipment type to demonstrate systematic multi-asset generalization. The distribution demonstrates comprehensive coverage across four equipment classes: bearings (32 scenarios across 8 datasets: CWRU, FEMTO, IMS, XJTU, HUST, MFPT, Mendeley, Paderborn), electric motors (8 scenarios across 2 datasets: ElectricMotorVibrations, RotorBrokenBar), gearboxes (5 scenarios across 2 datasets: GearboxUoC, PlanetaryPdM), and aircraft engines (30 scenarios across 6 datasets: CMAPSS FD001-004, Azure, EngineMTQA). This diversity ensures that agents are evaluated on their ability to generalize across equipment architectures, sensor modalities (vibration, current, temperature, pressure, RPM, torque), and operational contexts (varying speeds, loads, environmental conditions) rather than overfitting to a single asset class or data distribution.

The bearing dataset diversity merits particular emphasis, as it represents the most extensive multi-dataset coverage within a single equipment class. The eight bearing datasets offer complementary characteristics for comprehensive evaluation: CWRU provides standard fault types with controlled severity levels, FEMTO enables run-to-failure degradation analysis, IMS offers high-temporal-resolution vibration signatures (20kHz sampling), XJTU captures five distinct fault modes, HUST tests robustness across varying speeds (600-1500 RPM) and loads, MFPT enables frequency-domain analysis through ultra-high sampling rates (97.656 KHz), Mendeley tests non-stationary signal processing with varying speeds (600-1800 RPM), and Paderborn provides multimodal sensor fusion (vibration + current + torque + speed). This systematic diversity within bearings alone tests agent capabilities in cross-platform generalization—a critical requirement for industrial deployment where maintenance teams manage heterogeneous bearing types across different manufacturers, operational settings, and degradation modes.

Query Characteristics and Structural Diversity: Query characteristic statistics further reveal scenario diversity across dimensions that mirror real operational workflows. Of the 75 scenarios, approximately 60% (45 scenarios) are open-ended analytical questions requiring synthesized explanations ("What factors are contributing to premature bearing failures under high-load conditions in the HUST dataset?"), while 40% (30 scenarios) are close-ended questions with deterministic answers or multiple-choice options ("Does turbofan unit 23 meet FAA airworthiness requirements for continued operation—yes or no?"). This distribution reflects authentic information needs: sometimes stakeholders need exploratory analysis and diagnostic reasoning, while other times they need definitive determinations for compliance or operational decisions.

Data provision mode statistics demonstrate balanced coverage of two common industrial personas. Approximately 55% (41 scenarios) require agents to perform data discovery and analytical planning by identifying appropriate datasets (e.g., determining whether to use CMAPSS_FD001 or FEMTO_bearing_1 based on query context), loading relevant data files, and selecting analytical approaches. The remaining 45% (34 scenarios) embed necessary data within the query itself through sensor readings, operational parameters, or maintenance histories, focusing evaluation on diagnostic reasoning given complete information. This distinction represents operational manager personas (who pose high-level questions expecting autonomous data gathering) versus domain expert personas (who provide complete technical context in their queries).

Multi-asset fleet analysis comprises roughly 30% (23 scenarios) of the benchmark, requiring agents to process multiple equipment units simultaneously, compute predictions or classifications across the fleet, rank by risk or priority (typically lowest RUL = highest risk), apply operational threshold criteria (e.g., "identify units failing within 30 days"), and generate prioritized maintenance lists. The remaining 70% (52 scenarios) focus on single-asset analysis with deeper diagnostic detail on individual equipment instances. This 30/70 split reflects actual industrial workflows where fleet management occurs periodically (daily or weekly maintenance planning sessions) while diagnostic troubleshooting occurs continuously (responding to individual equipment alarms or anomalies).

Organizational Stakeholder Mapping: Organizational level mapping reveals that scenarios span all hierarchical levels of industrial decision-making, enabling evaluation of agent adaptability across diverse stakeholder needs. Approximately 35% of scenarios (26 scenarios) target site engineering and asset management levels, serving maintenance technicians (fault diagnosis with repair procedure recommendations), reliability engineers (RUL prediction for equipment prioritization), and condition monitoring specialists (sensor signal interpretation and anomaly detection). Approximately 40% (30 scenarios) target management and executive levels, serving plant managers (strategic maintenance planning and resource allocation), financial analysts (cost-benefit optimization and ROI justification), department directors (budget forecasting and capital expenditure planning), and vice presidents (portfolio-level risk assessment). The remaining 25% (19 scenarios) target regulatory and safety levels, serving environmental health & safety officers (compliance determination and safety protocol generation), compliance managers (multi-framework regulatory adherence), and risk assessors (four-level risk classification and mitigation strategy development).

This stakeholder distribution ensures that our benchmark evaluates agent capabilities in serving diverse industrial personas with heterogeneous priorities, vocabularies, and decision contexts. A maintenance technician needs immediate, actionable diagnostic information with component-level fault localization ("inner race damage at 0.014-inch crack depth, recommend bearing replacement within 48 hours"). A plant manager needs strategic analysis with business impact quantification ("preventive maintenance on 12 high-risk units costs $87k but avoids $340k in reactive failure costs and 156 hours of unplanned downtime"). A safety officer needs regulatory compliance verification with standard citations ("Unit 47 vibration exceeds OSHA PEL of 0.5g RMS, requires immediate operational restriction per 29 CFR 1910.95"). Our scenario design captures these distinct stakeholder-aligned task requirements through authentic vocabulary, appropriate analytical depth, and decision-relevant outputs.

Benchmark Evolution and Quality Investment: The complete scenario generation process, from initial proof-of-concept (1 scenario) through systematic expansion to comprehensive coverage (75 scenarios), represents approximately 396 person-hours of expert effort distributed across three specialized roles. This substantial human-intensive investment demonstrates our commitment to benchmark quality, real-world relevance, and evaluation rigor through progressive curation and expert annotation. A primary scenario writer with dual expertise in predictive maintenance and data science invested approximately 200 hours drafting technical scenarios, performing systematic dataset-task alignment, and defining ground truth validation criteria through empirical calibration against actual data distributions. Two SME reviewers contributed 196 hours combined (100 hours from an operations and maintenance leadership SME, 96 hours from an aerospace and manufacturing domain expert SME), validating real-world operational relevance, ensuring technical accuracy across diverse equipment types, and refining scenarios through multiple review iterations with detailed feedback.

This labor-intensive methodology, totaling $63,360 in expert effort at industry-standard overhead rates ($160/hour including benefits, facilities, management), ensures that every scenario reflects genuine operational needs encountered daily in industrial maintenance operations, uses authentic domain vocabulary with appropriate technical terminology density (averaging 4.2 technical terms per query across PHM acronyms, regulatory standards, equipment components, and diagnostic metrics), and can be objectively evaluated against verified ground truth through deterministic protocols rather than subjective assessment. While automated scenario generation methods could produce larger benchmarks more rapidly through LLM-based synthesis, our human-centric approach guarantees that each scenario represents actual industrial workflows, incorporates domain expertise accumulated through years of maintenance experience, and maintains evaluation validity through empirical ground truth rather than synthetic approximations.

The systematic expansion from 1 to 75 scenarios over a six-month development period (January to July 2025) demonstrates our deliberate growth strategy: validate evaluation framework through proof-of-concept (1 scenario, Week 1-2) → expand task diversity (10 scenarios, Week 3-6) → broaden strategic reasoning (20 scenarios, Week 7-12) → achieve multi-asset coverage (40 scenarios, Week 13-20) → incorporate multi-modal cognitive analysis (75 scenarios, Week 21-26). This progressive approach enabled iterative refinement of ground truth extraction protocols, query formulation guidelines, and evaluation criteria based on early validation results, ensuring benchmark quality scaled with coverage expansion.

4 Benchmark? - Ayan

Discuss relevant benchmarks here.

4.0.1 *MCP Servers.*

- discuss only definition of tools
- reference dhaval's paper, and use figure 1 for refernece, for tool writting.

4.1 Enterprise Level Agent Framework

- discuss **claude code**, **cursor code** and **ReAct**

4.2 LLMs

- discuss the different LLMs used, and their performance.

5 Experimental Study - Ayan

Explain what experiments was done, model's strengths and weaknesses, next steps and contributions encouragement here.

6 Conclusion - Ayan

This is the section to conclude the results of the research.

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Each author must be defined separately for accurate metadata identification. As an exception, multiple authors may share one affiliation. Authors' names should not be abbreviated; use full first names wherever possible. Include authors' e-mail addresses whenever possible.

Grouping authors' names or e-mail addresses, or providing an "e-mail alias," as shown below, is not acceptable:

```
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\email{dave,judy,steve@university.edu}
\email{firstname.lastname@phillips.org}
```

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```
\renewcommand{\shortauthors}{McCartney, et al.}
```

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The article template's documentation, available at https://www.acm.org/publications/proceedings-template, has a complete explanation of these commands and tips for their effective use.

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9 CCS Concepts and User-Defined Keywords

Two elements of the "acmart" document class provide powerful taxonomic tools for you to help readers find your work in an online search.

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User-defined keywords are a comma-separated list of words and phrases of the authors' choosing, providing a more flexible way of describing the research being presented.

CCS concepts and user-defined keywords are required for for all articles over two pages in length, and are optional for one- and two-page articles (or abstracts).

Table 1: Frequency of Special Characters

Non-English or Math	Frequency	Comments
Ø	1 in 1,000	For Swedish names
π	1 in 5	Common in math
$	4 in 5	Used in business
Ψ_1^2	1 in 40,000	Unexplained usage

10 Sectioning Commands

Your work should use standard LaTeX sectioning commands: `\section`, `\subsection`, `\subsubsection`, `\paragraph`, and `\subparagraph`. The sectioning levels up to `\subsusection` should be numbered; do not remove the numbering from the commands.

Simulating a sectioning command by setting the first word or words of a paragraph in boldface or italicized text is **not allowed.**

Below are examples of sectioning commands.

10.1 Subsection

This is a subsection.

10.1.1 Subsubsection. This is a subsubsection.

Paragraph. This is a paragraph.

Subparagraph This is a subparagraph.

11 Tables

The "`acmart`" document class includes the "`booktabs`" package — https://ctan.org/pkg/booktabs — for preparing high-quality tables.

Table captions are placed *above* the table.

Because tables cannot be split across pages, the best placement for them is typically the top of the page nearest their initial cite. To ensure this proper "floating" placement of tables, use the environment **table** to enclose the table's contents and the table caption. The contents of the table itself must go in the **tabular** environment, to be aligned properly in rows and columns, with the desired horizontal and vertical rules. Again, detailed instructions on **tabular** material are found in the *LaTeX User's Guide.*

Immediately following this sentence is the point at which Table 1 is included in the input file; compare the placement of the table here with the table in the printed output of this document.

To set a wider table, which takes up the whole width of the page's live area, use the environment **table*** to enclose the table's contents and the table caption. As with a single-column table, this wide table will "float" to a location deemed more desirable. Immediately following this sentence is the point at which Table 2 is included in the input file; again, it is instructive to compare the placement of the table here with the table in the printed output of this document.

Always use midrule to separate table header rows from data rows, and use it only for this purpose. This enables assistive technologies to recognise table headers and support their users in navigating tables more easily.

Table 2: Some Typical Commands

Command	A Number	Comments
`\author`	100	Author
`\table`	300	For tables
`\table*`	400	For wider tables

12 Math Equations

You may want to display math equations in three distinct styles: inline, numbered or non-numbered display. Each of the three are discussed in the next sections.

12.1 Inline (In-text) Equations

A formula that appears in the running text is called an inline or in-text formula. It is produced by the **math** environment, which can be invoked with the usual `\begin . . . \end` construction or with the short form `$ . . . $`. You can use any of the symbols and structures, from α to ω, available in LaTeX [25]; this section will simply show a few examples of in-text equations in context. Notice how this equation: $\lim_{n\rightarrow \infty}x=0$, set here in in-line math style, looks slightly different when set in display style. (See next section).

12.2 Display Equations

A numbered display equation—one set off by vertical space from the text and centered horizontally—is produced by the **equation** environment. An unnumbered display equation is produced by the **displaymath** environment.

Again, in either environment, you can use any of the symbols and structures available in LaTeX; this section will just give a couple of examples of display equations in context. First, consider the equation, shown as an inline equation above:

$$\lim_{n\rightarrow \infty}x=0 \tag{1}$$

Notice how it is formatted somewhat differently in the **displaymath** environment. Now, we'll enter an unnumbered equation:

$$\sum_{i=0}^{\infty} x + 1$$

and follow it with another numbered equation:

$$\sum_{i=0}^{\infty}x_i=\int_{0}^{\pi+2} f \tag{2}$$

just to demonstrate LaTeX's able handling of numbering.

13 Figures

The "`figure`" environment should be used for figures. One or more images can be placed within a figure. If your figure contains third-party material, you must clearly identify it as such, as shown in the example below.

Your figures should contain a caption which describes the figure to the reader.

Figure captions are placed *below* the figure.

Every figure should also have a figure description unless it is purely decorative. These descriptions convey what's in the image

Figure 5: 1907 Franklin Model D roadster. Photograph by Harris & Ewing, Inc. [Public domain], via Wikimedia Commons. (https://goo.gl/VLCRBB).

to someone who cannot see it. They are also used by search engine crawlers for indexing images, and when images cannot be loaded.

A figure description must be unformatted plain text less than 2000 characters long (including spaces). **Figure descriptions should not repeat the figure caption – their purpose is to capture important information that is not already provided in the caption or the main text of the paper.** For figures that convey important and complex new information, a short text description may not be adequate. More complex alternative descriptions can be placed in an appendix and referenced in a short figure description. For example, provide a data table capturing the information in a bar chart, or a structured list representing a graph. For additional information regarding how best to write figure descriptions and why doing this is so important, please see https://www.acm.org/publications/taps/describing-figures/.

13.1 The "Teaser Figure"

A "teaser figure" is an image, or set of images in one figure, that are placed after all author and affiliation information, and before the body of the article, spanning the page. If you wish to have such a figure in your article, place the command immediately before the `\maketitle` command:

```
\begin{teaserfigure}
  \includegraphics[width=\textwidth]{sampleteaser}
  \caption{figure caption}
```

```
  \Description{figure description}
\end{teaserfigure}
```

14 Citations and Bibliographies

The use of BibTeX for the preparation and formatting of one's references is strongly recommended. Authors' names should be complete — use full first names ("Donald E. Knuth") not initials ("D. E. Knuth") — and the salient identifying features of a reference should be included: title, year, volume, number, pages, article DOI, etc.

The bibliography is included in your source document with these two commands, placed just before the `\end{document}` command:

```
\bibliographystyle{ACM-Reference-Format}
\bibliography{bibfile}
```

where "`bibfile`" is the name, without the "`.bib`" suffix, of the BibTeX file.

Citations and references are numbered by default. A small number of ACM publications have citations and references formatted in the "author year" style; for these exceptions, please include this command in the **preamble** (before the command "`\begin{document}`") of your LaTeX source:

```
\citestyle{acmauthoryear}
```

Some examples. A paginated journal article [2], an enumerated journal article [11], a reference to an entire issue [10], a monograph (whole book) [24], a monograph/whole book in a series (see 2a in spec. document) [18], a divisible-book such as an anthology or compilation [13] followed by the same example, however we only output the series if the volume number is given [14] (so Editor00a's series should NOT be present since it has no vol. no.), a chapter in a divisible book [37], a chapter in a divisible book in a series [12], a multi-volume work as book [23], a couple of articles in a proceedings (of a conference, symposium, workshop for example) (paginated proceedings article) [3, 16], a proceedings article with all possible elements [36], an example of an enumerated proceedings article [15], an informally published work [17], a couple of preprints [6, 8], a doctoral dissertation [9], a master's thesis: [4], an online document / world wide web resource [1, 29, 38], a video game (Case 1) [28] and (Case 2) [27] and [26] and (Case 3) a patent [35], work accepted for publication [32], 'YYYYb'-test for prolific author [33] and [34]. Other cites might contain 'duplicate' DOI and URLs (some SIAM articles) [22]. Boris / Barbara Beeton: multi-volume works as books [20] and [19]. A presentation [31]. An article under review [7]. A couple of citations with DOIs: [21, 22]. Online citations: [38–40]. Artifacts: [30] and [5].

15 Acknowledgments

Identification of funding sources and other support, and thanks to individuals and groups that assisted in the research and the preparation of the work should be included in an acknowledgment section, which is placed just before the reference section in your document.

This section has a special environment:

```
\begin{acks}
...
\end{acks}
```

so that the information contained therein can be more easily collected during the article metadata extraction phase, and to ensure consistency in the spelling of the section heading.

Authors should not prepare this section as a numbered or unnumbered `\section`; please use the "`acks`" environment.

16 Appendices

If your work needs an appendix, add it before the "`\end{document}`" command at the conclusion of your source document.

Start the appendix with the "`appendix`" command:

```
\appendix
```

and note that in the appendix, sections are lettered, not numbered. This document has two appendices, demonstrating the section and subsection identification method.

17 Multi-language papers

Papers may be written in languages other than English or include titles, subtitles, keywords and abstracts in different languages (as a rule, a paper in a language other than English should include an English title and an English abstract). Use `language=...` for every language used in the paper. The last language indicated is the main language of the paper. For example, a French paper with additional titles and abstracts in English and German may start with the following command

```
\documentclass[sigconf, language=english, language=german,
               language=french]{acmart}
```

The title, subtitle, keywords and abstract will be typeset in the main language of the paper. The commands `\translatedXXX`, `XXX` begin title, subtitle and keywords, can be used to set these elements in the other languages. The environment `translatedabstract` is used to set the translation of the abstract. These commands and environment have a mandatory first argument: the language of the second argument. See `sample-sigconf-i13n.tex` file for examples of their usage.

18 SIGCHI Extended Abstracts

The "`sigchi-a`" template style (available only in LaTeX and not in Word) produces a landscape-orientation formatted article, with a wide left margin. Three environments are available for use with the "`sigchi-a`" template style, and produce formatted output in the margin:

`sidebar`: Place formatted text in the margin.
`marginfigure`: Place a figure in the margin.
`margintable`: Place a table in the margin.

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References

[1] Rafal Ablamowicz and Bertfried Fauser. 2007. *CLIFFORD: a Maple 11 Package for Clifford Algebra Computations, version 11.* Retrieved February 28, 2008 from http://math.tntech.edu/rafal/cliff11/index.html
[2] Patricia S. Abril and Robert Plant. 2007. The patent holder's dilemma: Buy, sell, or troll? *Commun. ACM* 50, 1 (Jan. 2007), 36–44. doi:10.1145/1188913.1188915
[3] Sten Andler. 1979. Predicate Path expressions. In *Proceedings of the 6th. ACM SIGACT-SIGPLAN symposium on Principles of Programming Languages (POPL '79).* ACM Press, New York, NY, 226–236. doi:10.1145/567752.567774

[4] David A. Anisi. 2003. *Optimal Motion Control of a Ground Vehicle*. Master's thesis. Royal Institute of Technology (KTH), Stockholm, Sweden.
[5] Sam Anzaroot and Andrew McCallum. 2013. *UMass Citation Field Extraction Dataset*. Retrieved May 27, 2019 from http://www.iesl.cs.umass.edu/data/data-umasscitationfield
[6] Sam Anzaroot, Alexandre Passos, David Belanger, and Andrew McCallum. 2014. *Learning Soft Linear Constraints with Application to Citation Field Extraction*. arXiv:1403.1349 doi:10.48550/arXiv.1403.1349
[7] R. Baggett, M. Simecek, C. Chambellan, K. Tsui, and M. Fraune. 2025. Fluidity in the Phased Framework of Technology Acceptance: Case Study to Gain a Holistic Understanding of (Older Adult) Participant Advancement Through Acceptance Phases with Mobile Telepresence Robots. *Robotics Aut. Systems*. Manuscript submitted for review.
[8] Lutz Bornmann, K. Brad Wray, and Robin Haunschild. 2019. *Citation concept analysis (CCA)—A new form of citation analysis revealing the usefulness of concepts for other researchers illustrated by two exemplary case studies including classic books by Thomas S. Kuhn and Karl R. Popper*. arXiv:1905.12410 [cs.DL]
[9] Kenneth L. Clarkson. 1985. *Algorithms for Closest-Point Problems (Computational Geometry)*. Ph. D. Dissertation. Stanford University, Palo Alto, CA. UMI Order Number: AAT 8506171.
[10] Jacques Cohen (Ed.). 1996. Special issue: Digital Libraries. *Commun. ACM* 39, 11 (Nov. 1996).
[11] Sarah Cohen, Werner Nutt, and Yehoshua Sagic. 2007. Deciding equivalances among conjunctive aggregate queries. *J. ACM* 54, 2, Article 5 (April 2007), 50 pages. doi:10.1145/1219092.1219093
[12] Bruce P. Douglass, David Harel, and Mark B. Trakhtenbrot. 1998. Statecarts in use: structured analysis and object-orientation. In *Lectures on Embedded Systems*, Grzegorz Rozenberg and Frits W. Vaandrager (Eds.). Lecture Notes in Computer Science, Vol. 1494. Springer-Verlag, London, 368–394. doi:10.1007/3-540-65193-4_29
[13] Ian Editor (Ed.). 2007. *The title of book one* (1st. ed.). The name of the series one, Vol. 9. University of Chicago Press, Chicago, Chapter The title of the chapter, 127–238. doi:10.1007/3-540-09237-4
[14] Ian Editor (Ed.). 2008. *The title of book two* (2nd. ed.). University of Chicago Press, Chicago, Chapter 100, 25–137. doi:10.1007/3-540-09237-4
[15] Matthew Van Gundy, Davide Balzarotti, and Giovanni Vigna. 2007. Catch me, if you can: Evading network signatures with web-based polymorphic worms. In *Proceedings of the first USENIX workshop on Offensive Technologies (WOOT '07)*. USENIX Association, Berkley, CA, Article 7, 9 pages.
[16] Torben Hagerup, Kurt Mehlhorn, and J. Ian Munro. 1993. Maintaining Discrete Probability Distributions Optimally. In *Proceedings of the 20th International Colloquium on Automata, Languages and Programming (Lecture Notes in Computer Science, Vol. 700)*. Springer-Verlag, Berlin, 253–264.
[17] David Harel. 1978. *LOGICS of Programs: AXIOMATICS and DESCRIPTIVE POWER*. MIT Research Lab Technical Report TR-200. Massachusetts Institute of Technology, Cambridge, MA.
[18] David Harel. 1979. *First-Order Dynamic Logic*. Lecture Notes in Computer Science, Vol. 68. Springer-Verlag, New York, NY. doi:10.1007/3-540-09237-4
[19] Lars Hörmander. 1985. *The analysis of linear partial differential operators. III*. Grundlehren der Mathematischen Wissenschaften [Fundamental Principles of Mathematical Sciences], Vol. 275. Springer-Verlag, Berlin, Germany. viii+525 pages. Pseudodifferential operators.
[20] Lars Hörmander. 1985. *The analysis of linear partial differential operators. IV*. Grundlehren der Mathematischen Wissenschaften [Fundamental Principles of Mathematical Sciences], Vol. 275. Springer-Verlag, Berlin, Germany. vii+352 pages. Fourier integral operators.
[21] IEEE 2004. IEEE TCSC Executive Committee. In *Proceedings of the IEEE International Conference on Web Services (ICWS '04)*. IEEE Computer Society, Washington, DC, USA, 21–22. doi:10.1109/ICWS.2004.64
[22] Markus Kirschmer and John Voight. 2010. Algorithmic Enumeration of Ideal Classes for Quaternion Orders. *SIAM J. Comput.* 39, 5 (Jan. 2010), 1714–1747. doi:10.1137/080734467
[23] Donald E. Knuth. 1997. *The Art of Computer Programming, Vol. 1: Fundamental Algorithms (3rd. ed.)*. Addison Wesley Longman Publishing Co., Inc., Boston.
[24] David Kosiur. 2001. *Understanding Policy-Based Networking* (2nd. ed.). Wiley, New York, NY.
[25] Leslie Lamport. 1986. *LaTeX: A Document Preparation System*. Addison-Wesley, Reading, MA.
[26] Newton Lee. 2005. Interview with Bill Kinder: January 13, 2005. Video. *Comput. Entertain.* 3, 1, Article 4 (Jan.-March 2005). doi:10.1145/1057270.1057278
[27] Dave Novak. 2003. Solder man. Video. In *ACM SIGGRAPH 2003 Video Review on Animation theater Program: Part I - Vol. 145 (July 27–27, 2003)*. ACM Press, New York, NY, 4. doi:10.945/woot07-S422 http://video.google.com/videoplay?docid=6528042696351994555
[28] Barack Obama. 2008. A more perfect union. Video. Retrieved March 21, 2008 from http://video.google.com/videoplay?docid=6528042696351994555
[29] Poker-Edge.Com. 2006. Stats and Analysis. Retrieved June 7, 2006 from http://www.poker-edge.com/stats.php
[30] R Core Team. 2019. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria. https://www.R-project.org/
[31] Brian J. Reiser. 2014. Designing coherent storylines aligned with NGSS for the K-12 classroom. Presentation at National Science Education Leadership Association Meeting, Boston, MA, USA. https://www.academia.edu/6884962/
[32] Bernard Rous. 2008. The Enabling of Digital Libraries. *Digital Libraries* 12, 3, Article 5 (July 2008). To appear.
[33] Mehdi Saeedi, Morteza Saheb Zamani, and Mehdi Sedighi. 2010. A library-based synthesis methodology for reversible logic. *Microelectron. J.* 41, 4 (April 2010), 185–194.
[34] Mehdi Saeedi, Morteza Saheb Zamani, Mehdi Sedighi, and Zahra Sasanian. 2010. Synthesis of Reversible Circuit Using Cycle-Based Approach. *J. Emerg. Technol. Comput. Syst.* 6, 4 (Dec. 2010), 12 pages.
[35] Joseph Scientist. 2009. The fountain of youth. Patent No. 12345, Filed July 1st., 2008, Issued Aug. 9th., 2009.
[36] Stan W. Smith. 2010. An experiment in bibliographic mark-up: Parsing metadata for XML export. In *Proceedings of the 3rd. annual workshop on Librarians and Computers (LAC '10, Vol. 3)*, Reginald N. Smythe and Alexander Noble (Eds.). Paparazzi Press, Milan Italy, 422–431.
[37] Asad Z. Spector. 1990. Achieving application requirements. In *Distributed Systems* (2nd. ed.), Sape Mullender (Ed.). ACM Press, New York, NY, 19–33. doi:10.1145/90417.90738
[38] Harry Thornburg. 2001. *Introduction to Bayesian Statistics*. Retrieved March 2, 2005 from http://ccrma.stanford.edu/~jos/bayes/bayes.html, archived at [https://web.archive.org/web/20240505055615/https://ccrma.stanford.edu/~jos/bayes/bayes.html]
[39] TUG 2017. *Institutional members of the TeX Users Group*. Retrieved May 27, 2017 from http://wwtug.org/instmem.html
[40] Boris Veytsman. 2017. *acmart—Class for typesetting publications of ACM*. Retrieved May 27, 2017 from http://www.ctan.org/pkg/acmart

A Research Methods

A.1 Part One

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